



USN

--	--	--	--	--	--	--	--	--	--

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Seventh Semester B.Tech (CBCS) Degree Examinations

December 2025

BT2201-1 – BASICS OF PHARMACEUTICAL SCIENCE

(For BT)

Duration: 3 Hours

Max. Marks:100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Fivefull** questions choosing **Two full** question from **Unit – I & II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. What is a key focus of pharmaceutical biotechnology?
A) Herbal medicine production B) Patient counseling
C) Drug design and development D) Hospital management
2. For what purpose are orphan drugs primarily used?
A) Common cold B) Hypertension
C) Rare diseases D) Diabetes
3. The ultimate goal of pharmacovigilance is:
A) Accelerating production B) Improving marketing strategies
C) Reducing manufacturing costs D) Protecting patients from unsafe medicines
4. How can biotherapeutics be best described?
A) Synthetic chemicals B) Nutritional supplements
C) Generic tablets D) Biological molecules used as drugs
5. Identify the class of drugs that sulpha drugs belong to:
A) Antibiotics B) Antihistamines
C) Analgesics D) Hormones
6. Which of the following defines nutraceuticals?
A) Prescription drugs only B) Food-derived products with health benefits
C) Vaccines D) Synthetic steroids
7. In what way do generic drugs primarily differ from brand-name drugs?
A) Chemical structure B) Bioavailability
C) Price and branding D) Therapeutic action
8. State the components included in biotherapeutics:
A) Monoclonal antibodies and recombinant proteins B) Over-the-counter painkillers
C) Sterile water D) Hormonal contraceptives
9. In ADME, what does 'A' stand for?
A) Administration B) Absorption
C) Affinity D) Activation
10. How does passive diffusion of drugs occur?
A) Energy expenditure by cells B) Concentration gradient
C) Carrier proteins D) Endocytosis
11. What factor affects drug absorption the most?
A) Drug solubility B) Drug price
C) Color of the tablet D) Prescribing doctor
12. Which process is primarily involved in Phase I metabolism?
A) Conjugation reactions B) Oxidation, reduction, hydrolysis
C) Gastrointestinal absorption D) Renal excretion

13. Which method is commonly used in the manufacture of ointment bases?
 A) Incorporation of drug into molten base or fusion of ingredients
 B) Lyophilization of the active ingredient
 C) Direct compression into tablets
 D) Solvent evaporation to form tablets
14. In what way do hard gelatin capsules differ from soft gelatin capsules?
 A) Drug solubility
 B) Shell composition
 C) Therapeutic effect
 D) Bioavailability
15. Which dosage form bypasses first-pass metabolism?
 A) Oral tablets
 B) Capsules
 C) Sublingual tablets
 D) Powders
16. How are compressed tablets typically manufactured?
 A) Lyophilization
 B) Extrusion
 C) Direct compression
 D) Spray drying
17. What does absolute bioavailability compare?
 A) Two different drugs
 B) Single dose vs multiple dose
 C) Tablet vs capsule
 D) Oral vs intravenous administration of the same drug
18. Identify the meaning of IVIVC:
 A) In vivo-in vitro correlation
 B) Intravenous infusion volume calculation
 C) Immediate versus delayed drug clearance
 D) Ionic voltage interaction coefficient
19. Why are controlled-release oral dosage forms designed?
 A) Release drug rapidly
 B) Avoid absorption
 C) Increase drug toxicity
 D) Maintain therapeutic concentration over time
20. Which route of drug delivery primarily targets the lungs?
 A) Oral
 B) Transdermal
 C) Intravenous
 D) Inhalation

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Outline and describe the stages of drug discovery and development, from initial discovery to market approval.	8	L*2	1	1,6
	b) List the types of laxatives and explain their mechanisms of action along with their clinical uses.	8	L1	2	2,4
2.	a) Classify prescription and non-prescription drugs, and analyze how drug abuse and dependence impact society.	8	L2	1	1,6
	b) Illustrate the indications for hormone replacement therapy and interpret its benefits.	8	L2	2	2,4
3.	a) List the different types of approved biotherapeutics, explain their mechanisms, and provide examples of their clinical applications.	8	L1	2	2,4
	b) Outline the scope of pharmaceutical biotechnology in modern healthcare.	8	L2	1	1,6
Unit – II					
4.	a) How does ion-pair transport facilitate drug absorption across biological membranes?	8	L1	3	1,3
	b) Illustrate the tablet coating process, including its stages, classification, and importance in pharmaceutical manufacturing.	8	L2	4	1,6
5.	a) List patient-related factors influencing drug absorption and show how each factor affects bioavailability.	8	L1	3	1,3
	b) Outline the steps involved in the wet granulation process and explain its significance in tablet formulation.	8	L2	4	1,6

BT2201-1

SEE - December 2025

- | | | | | | |
|-------|---|---|----|---|-----|
| 6. a) | Compare zero-order, first-order, and mixed-order kinetics with suitable examples. | 8 | L2 | 4 | 1,6 |
| b) | Illustrate oxidative, reductive, and hydrolytic reactions in Phase I drug metabolism. | 8 | L2 | 3 | 1,3 |

Unit – III

- | | | | | | |
|-------|---|---|----|---|-----|
| 7. a) | Illustrate the methods to enhance drug bioavailability through improved membrane permeability and gastrointestinal retention. | 8 | L2 | 5 | 1,3 |
| b) | Illustrate intranasal controlled-release systems and explain their advantages and limitations. | 8 | L2 | 5 | 1,3 |
| 8. a) | List and explain the biological factors that influence tissue permeability of drugs, providing suitable examples for each factor. | 8 | L1 | 5 | 1,3 |
| b) | Classify oral controlled-release drug delivery systems with suitable examples. | 8 | L2 | 5 | 1,3 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
